

APPH3100: Introduction to Quantum Mechanics

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Abstract

The course starts with a recap of linear algebra and quantum mechanics, followed by an introduction to quantum optics and quantum information. Two-level systems, Bloch sphere, quantum gates, and elementary quantum algorithms will be discussed. Quantum teleportation and quantum key distribution will be introduced as applications of entanglement. The lecture content will be directly applied in experiments with entangled photons. In the following, state-of-the-art quantum algorithms will be discussed, related to cutting-edge research results in quantum computing. This includes quantum Fourier transform, quantum simulation of the Schroedinger equation, and the variational quantum eigensolver (VQE) algorithm. During the course students will do one experimental project with entangled photons and one quantum programming project. Students will be guided to implement a quantum algorithm of their choice and run it on a quantum computer (IBM, IonQ, QuEra).

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Chapter 1

Quantum Systems

Lecture 1

1.1 Quantum Computing Systems

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Example. Atoms: There is a nucleus denoted Ze with an electron denoted as e^- . Having a positively charged nucleus and electron can be described by a hamiltonian denoted as

$$H^{at} = \frac{p^2}{2m_e} + V(r), V(r) = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

Where the hamiltonian represented the movement of the electron.

In this potential, we get discrete energy states. More precisely, we can solve the schrodinger equation denoted

$$H^{at}\phi(\vec{r}) = E_n\phi(\vec{r})$$

The eigen energies have the form

$$E_n = -\frac{1}{2}m_e c^2 \frac{\alpha^2}{n^2}$$

This has discrete energy values and the smallest energy is for $n=1$ Where $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}$

The eigenfunctions have the form of the following:

$$\phi(\vec{r}) = R_{ne}(r)Y_{em}(\theta, \phi)$$

which includes the spherical harmonics. These eigenfunctions are also orthonormal wave functions. There are also the three quantum numbers denoted as: n, l , and m which are the principal quantum number, angular momentum, and projection angular momentum.

Intuitively for the hydrogen model, we see that the energystates come close and close as a factor of $\frac{1}{n^2}$

Remark. Simplification: We only use a two level system denoted as E_1, E_2 with wavfunction denoted as ψ_{100}, ψ_{210} which are going to be known as the s-orbital and the p-orbital which is a two level system.

Example. Spin $-\frac{1}{2}$: These systems include electrons, protons, and neutrons which have spins which are an intrinsic form of angular momentum. All these are spin $\frac{1}{2}$ particles. Spin always comes with a magnetic moment. One example is a electron and it has two states which are denoted as

$$|\uparrow\rangle, |\downarrow\rangle$$

These are orthogonal to each other with angular momentums $\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$. These electrons are in a

magnetic field pointing in the z direction $B = B\hat{e}_Z$

The interaction is denoted as

$$\hat{H}_z = \vec{\mu} \cdot \vec{B} = -(-g_s\mu_B\vec{S}/\hbar) \cdot B$$

where $\mu_B = \frac{e\hbar}{2m_e}$ which is the Bohr magneton. $g_s = 2$ which is known as the g-factor. The spin operator can be denoted in the following way

$$\vec{S} = (S_x, S_y, S_z)^T$$

which is simply $\frac{\hbar}{2}$ multiplied with the Pauli matrices. We can rewrite this as

$$g_s\mu_B \frac{S_z}{\hbar} B_z = \mu_B B_z \sigma_z$$

Thus we get that

$$H_z |\uparrow\rangle = \mu_B B_z |\uparrow\rangle$$

and negative for the latter.

With a 0 magnetic field, the spin up and spin down are the same. When you increase the energy, they have different states denoted as $\mu_B B_z$ and $-\mu_B B_z$. What you get here is called level splitting

Example. Photon Polarization: the electric field of a light wave in the z-direction may be in the form where the wave propagates in the z-direction. This is going to be polarized in the plane of the chalk board. This can be written as

$$\vec{E}(z, y) = (E_x, E_y e^{i\theta}) e^{i(kz - \omega t)}$$

where the values are known to be the angular frequency, wavelength, and wave vector. We can rewrite this polarization vector as

$$E_x \hat{e}_x + E_y e^{i\theta} \hat{e}_y$$

with $E_0 = \sqrt{E_x^2 + E_y^2}$. Thus it could have both an x and y component. If we have that $\theta = 0$, $E_y = 0 \Rightarrow E_x = E_0$ which is denoted as vertical linear polarization and the latter is called horizontal linear polarization if $E_y = E_0$. Each photon has its own unique polarization!

Polarization is a property carried by the individual photons that constitute the light field.

$$\vec{V} = E_0 \hat{e}_x, \vec{H} = E_0 \hat{e}_y$$

The polarization state vector of a single photon can be written as a superposition of $|H\rangle + e^{-i\phi} |V\rangle$ which is a two level system

Example. Superconducting Qubits: Suppose we have a LC-circuit with an inductor and a capacitor which creates a resonance with a frequency with angular frequency $\omega_c = \frac{1}{\sqrt{2C}}$

Suppose we super cool it down such that the wire becomes superconducting and for the case where $k_B T \ll \hbar\omega_c$ is much smaller than the excitation frequency of the circuit, we get discrete energy states for the oscillation that correspond to a quantum harmonic oscillator. We can write the hamiltonian of the circuit in the form

$$\hat{H} = \hat{q}^2/2C + \phi^2/2C$$

which is similar to the momentum and position operator we see in a quantum harmonic oscillator. $\hat{\phi}$ is the magnetic flux operator, $\hat{q}$ is the charge operator.

This creates a quadratic potential well with energy states $E_0 = \hbar\frac{\omega}{2}$ with spaces of $\hbar\omega$. The energies thus can be expressed as

$$E_n = \hbar\omega_c(n + \frac{1}{2})$$

If you want this to be a two level system, it is practically bad since you might couple to many other

states due to its equal spacings. We can replace the inductor l with a Josephson junction (quantum tunneling/nonlinear inductor) denoted as L_g . The hamiltonian of this system is not modified as

$$\hat{H} = \frac{q^2}{2C} - E_J \cos \phi$$

which is an anharmonic oscillator with energy levels that are different with spacings that get smaller and smaller as it goes up. This is useful for defining a two level system practical for control purposes where the energy levels $\omega_{21} < \omega_{10} \approx 2\pi \cdot 5GHz$ which can be controlled by microwaves. Thus we get a two level system!

Remark. Linear Algebra Review:

Definition 1.1.1. wave functions can be interpreted as vectors in a vector space V which is a hilbert space. We denote these in Dirac notation by $|\psi\rangle$. For linear combinations. For $c_i \in \mathbb{C}$ and $|\psi_i\rangle \in V$ we have that

$$|\psi\rangle = c_1 |\psi_1\rangle + \dots + c_n |\psi_n\rangle \in V$$

A set of vectors $|\psi\rangle$ is linearly independent if none of the vectors can be written as a linear combination of the other vectors. In addition, if such a set spans the entire vector space, it is called a basis.

Definition 1.1.2. Two level systems are described by 2 dimensional vector spaces in $\mathbb{C}^2$ spanned by two basis vectors $|v_1\rangle |v_2\rangle$. Any vector in that vector space can be expressed as

$$|\psi\rangle = c_1 |v_1\rangle + c_2 |v_2\rangle$$

Column vectors can be written as

$$|\psi\rangle = (c_1 c_2)^T$$

and the complex conjugate can be written as

$$|\psi\rangle = (c_1 c_2)^*$$

The scalar product is denoted as

$$\langle\beta|\alpha\rangle = (b_1^* b_2^*)(a_1 a_2)^T$$

Lecture 2

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Chapter 2

Schrodinger's Equation

2.1 Time Independent Equation

Remark. The time-independent Schrödinger equation is

$$\hat{H}\psi(x) = E\psi(x),$$

and for stationary states the probability density is time independent:

$$\frac{d}{dt} |\psi(x, t)|^2 = 0.$$

For the case $V(x) = 0$, a solution is the plane wave

$$\phi(x) = e^{ikx}.$$

Since

$$E = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m} \quad \text{with } p = \hbar k,$$

this solution represents a free particle. However, note that the plane wave is not normalizable (its probability density does not integrate to 1 over all space). To construct a physical (normalizable) state, we use a superposition of plane waves:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \gamma(k) e^{ikx} dk,$$

where $\gamma(k)$ is a weight function (analogous to a set of coefficients c_k in a discrete expansion) chosen so that $\phi(x)$ is square-integrable.

Remark. We now choose a specific form for $\gamma(k)$; for example, a Gaussian centered at k_0 :

$$\gamma(k) = A \exp\left[-\frac{a^2(k - k_0)^2}{4}\right].$$

The Fourier transform yields

$$\phi(x) = \frac{\sqrt{2}A}{a} \exp\left[-\frac{x^2}{a^2}\right] e^{ik_0x}.$$

Here, e^{ik_0x} represents a plane wave, while the Gaussian envelope $\exp(-x^2/a^2)$ determines the localization in position space. Note that a narrow Gaussian in momentum space (i.e. a well-defined momentum) corresponds to a wide Gaussian in position space. In this case the uncertainties satisfy

$$\Delta x \Delta k = \frac{1}{2}.$$

Including the time dependence (recalling that for each plane wave e^{ikx} we have

$$\phi(t) \propto e^{-i\omega t}, \quad \text{with } \omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m},$$

we can write the full time-dependent wave function.

Remark. Suppose we wish to prepare a particle in a wave packet with a momentum distribution sharply peaked around k_0 . In momentum space, this means

$$\frac{1}{a} \ll k_0.$$

We then Taylor expand the dispersion relation $\omega(k)$ about k_0 :

$$\omega(k) \approx \omega(k_0) + \left. \frac{d\omega}{dk} \right|_{k=k_0} (k - k_0).$$

Writing $k = k_0 + \delta$, the wave function becomes

$$\phi(x, t) = \frac{1}{\sqrt{2\pi}} e^{ik_0 x} e^{-i\omega(k_0)t} \int_{-\infty}^{\infty} \gamma(k_0 + \delta) e^{i\delta \left(x - \left. \frac{d\omega}{dk} \right|_{k=k_0} t \right)} d\delta.$$

This shows that there are two relevant velocities:

- The **phase velocity**:

$$v_{pw} = \frac{\omega(k_0)}{k_0} = \frac{\hbar k_0}{2m} = \frac{p}{2m},$$

which is the velocity associated with the phase of the carrier wave.

- The **group velocity**:

$$v_g = \left. \frac{d\omega}{dk} \right|_{k=k_0} = \frac{\hbar k_0}{m} = \frac{p}{m},$$

which is the velocity at which the envelope (and thus the particle) propagates.

Thus, although the carrier wave has a phase velocity of $p/(2m)$, the actual particle travels at the group (or classical) velocity p/m . In addition, if the momentum spread is not extremely narrow, the wave packet will spread over time.

Remark. Now consider a situation with a potential $V(x)$ such that at some point the energy E of the particle equals the potential, meaning the kinetic energy is zero there. These are the classical turning points where $E = V(x)$. However, the quantum mechanical wave function can “leak” (tunnel) beyond these classical turning points.

For example, consider a step potential defined by

$$V(x) = \begin{cases} 0, & x < a, \\ V_0, & x \geq a. \end{cases}$$

We label region 1 (where $x < a$) and region 2 (where $x \geq a$). Suppose that in region 1 $E > 0$ so that the solution is oscillatory:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi,$$

with the solution

$$\psi_1(x) = e^{ikx}, \quad \text{where } k = \sqrt{\frac{2mE}{\hbar^2}}.$$

In region 2 the Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V_0 \psi = E\psi.$$

Rearranging, we have

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = (E - V_0)\psi.$$

If $E < V_0$, then $E - V_0$ is negative; defining

$$\kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}},$$

the differential equation becomes

$$\frac{d^2\psi}{dx^2} = \kappa^2\psi,$$

with general solution

$$\psi_2(x) = A e^{-\kappa x} + B e^{\kappa x}.$$

For a physically acceptable solution (one that does not blow up as $x \rightarrow \infty$), we choose the decaying exponential (set $B = 0$). Notice that while the wave function $\psi(x)$ remains continuous across the boundary at $x = a$, its derivative may be discontinuous because the potential is discontinuous there.

Lecture 3

2.2 Infinite Square Well

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Remark. We will solve for a case called the infinite square well. We can write the potential to be

$$V(x) = \begin{cases} \infty, & \text{if } x < 0; \\ \infty, & \text{if } x > a; \\ 0, & \text{otherwise.} \end{cases}$$

$$E\psi = \frac{-\hbar^2}{2m} \left[\frac{d^2\psi}{dx^2} \right]$$

The solutions we have here are going to be

$$A_0 = e^{\pm ikx}, k = \sqrt{\frac{2mE}{\hbar^2}}$$

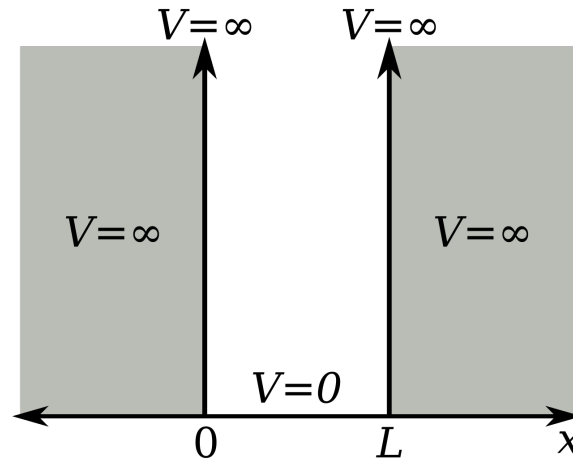


Figure 2.1

Recall that we have the equation

$$\begin{aligned}\psi &= Ae^{\alpha x} \\ -[A\alpha^2 e^{\alpha x}] \frac{\hbar^2}{2m} &= Ex \\ -\frac{\hbar^2 \alpha^2}{2m} &= E \\ \alpha^2 &= \frac{2mE}{\hbar^2} \rightarrow \alpha = \pm i \sqrt{\frac{2mE}{\hbar^2}}\end{aligned}$$

Remark. Second Order Differential Equation:

$$\psi = A \sin(kx) + B \cos(kx)$$

Note that $\sin(kx) = \frac{e^{ikx} - e^{-ikx}}{2}$ and $\cos(kx)$ to have a positive term. We notice that immediately that the $\cos(kx)$ term cannot be in the solution since the potential is $V(0) = 0$.

$$\psi = A \sin(kx)$$

with an additional constraint that

$$\psi(x=a) = 0$$

$$A \sin(ka) = 0$$

$$ka = n\pi, k = \frac{n\pi}{a}$$

There are only quantized values of k which means that the energy is also quantized where

$$E = \frac{\hbar^2 k^2}{2m}$$

$$E_n = \frac{\hbar^2 \pi^2}{2ma^2} n^2$$

Thus quantized energy states come from boundary conditions which lead to quantized energies. For free particles, energy is continuous but as we put boundary conditions, energy is discretized.

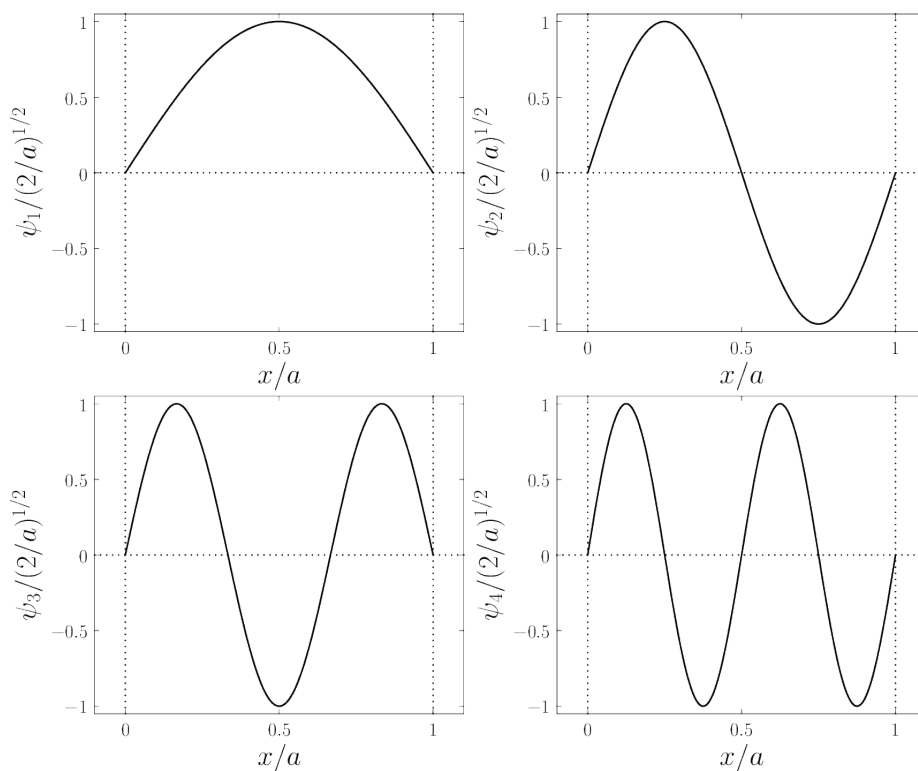


Figure 2.2

These are the solutions for $n = 1, 2, 3, 4$ where we see an additional oscillation for each term.

Example. For a given example, suppose we had a potential with 6 zeroes for our ψ , then our energy can be approximated to be

$$E \approx \frac{\hbar^2 \pi^2}{2ma^2} \times 5^2$$



Figure 2.3

Remark. Notice that we have been carrying a value A , we must normalize to calculate expectation

values. We normalize through the L2 norm which is given by

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

One nice thing about the infinite square well is that it is finite. thus we have

$$\int_0^a A_n^2 \sin^2\left(\pi \frac{n}{a} x\right) dx = 1$$

We can redefine the argument and rescale giving a unitless integral.

$$y = \pi \frac{n}{a} x, dy = \pi \frac{n}{a} dx,$$

where y is a unitless term.

$$\begin{aligned} A_n^2 \frac{a}{n\pi} \int_0^{n\pi} \sin^2(y) dy &= 1 \\ A_n^2 \frac{a}{n\pi} \left[\frac{1}{2} (y - \sin(y) \cos(y)) \right] \Big|_0^{n\pi} &= 1 \\ A_n^2 \frac{a}{n\pi} \frac{1}{2} n\pi &= 1 \\ A_n^2 &= \frac{2}{a} \\ A_n &= \sqrt{\frac{2}{a}} \end{aligned}$$

Thus, all normalized wave functions look like the following:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\pi \frac{nx}{a}\right)$$

Notice that the time dependence is simply a separation of variables where

$$\Psi(x, t) = \psi_n(x) \phi_n(t)$$

where

$$\phi_n(t) = e^{-iE_n \frac{t}{\hbar}}$$

Any expectation value we look at will be invariant with time. Notice that

$$e^{-i \frac{E_n}{\hbar} t} = \cos\left(\frac{E_n}{\hbar} t\right) - i \sin\left(\frac{E_n}{\hbar} t\right)$$

Remark. Superposition:

$$\psi(x, t) = \frac{1}{\sqrt{2}} \sqrt{\frac{2}{a}} \left[\sin\left(\pi \frac{x}{a}\right) + \sin\left(2\pi \frac{x}{a}\right) \right]$$

This is the solution at $t = 0$ where it is the superposition of the $n = 1$ and $n = 2$ solution. Suppose that we did not have the $\frac{1}{\sqrt{2}}$ part, we would need to normalize.

$$\psi = C \sqrt{\frac{2}{a}} [\psi_1 + \psi_2]$$

which is an equally weighted superposition. We can normalize now with

$$\int_0^a |\psi|^2 dx = |C|^2 \frac{2}{a} \int_0^a [|\psi_1|^2 + |\psi_2|^2 + \psi_1^* \psi_2 + \psi_1 \psi_2^*] dx$$

The only contribution will come from the absolute value terms. Coming back we can write

$$\psi_n(x, t) = \frac{1}{\sqrt{a}} \left[\psi_1(x) e^{-\frac{iE_1}{\hbar} t} + \psi_2(x) e^{-\frac{iE_2}{\hbar} t} \right]$$

We will pick up when we have two stationary states in a infinite square well. We will end up getting some in phase and out of phase which will lead to varying time dependent argument presented here.

Lecture 4

2.3 Infintie Potential Square Well

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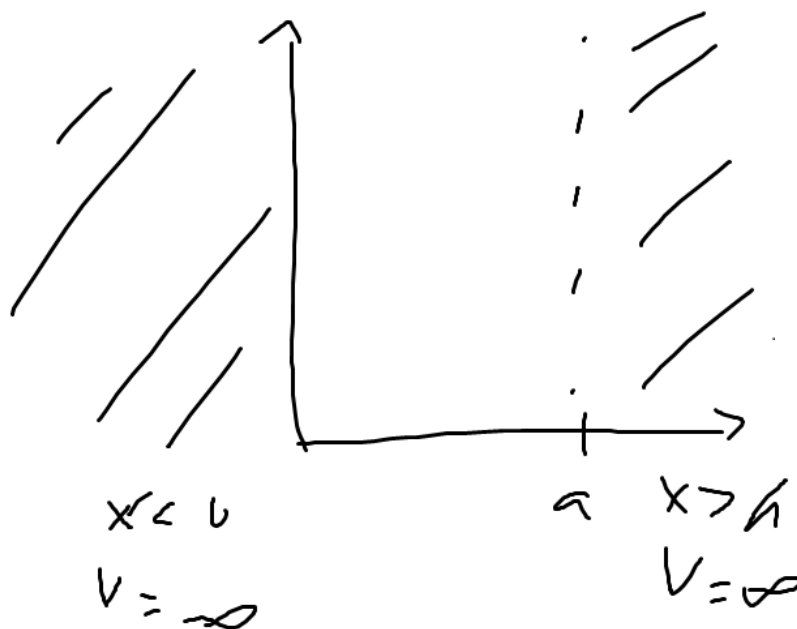


Figure 2.4

Remark. Classically what we would see is a particle going back and forth in the square well. If we take a snapshot in time, there would be an equal probability being anywhere (not really but roughly speaking). In the classic case, we recall that the energy states of the first energy level was

$$\psi = \sin\left(\pi \frac{x}{a}\right)$$

as shown in Figure 1.2 and the probability would be

$$|\psi_1|^2 = \alpha^2 \sin^2\left(\pi \frac{x}{a}\right)$$

The next solution we have is where $n = 2$, $\psi = \alpha \sin\left(2\pi \frac{x}{a}\right)$ and we can observe where the function

goes to 0.

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right), A = \sqrt{\frac{2}{a}}$$

$$1 = A^2 \int_0^a dx |\psi_n(x)|^2$$

where we can determine the normalization factor from here.

Remark. The superposition we want to consider now is

$$\psi(x, 0) = \frac{1}{\sqrt{2}} [\psi_1 + \psi_2]$$

Now we are determining the expectation value of the energy $\langle E \rangle$

$$\int_0^x \psi^* \hat{H} \psi dx$$

note that applying the Hamiltonian operator is applied on the real operator as when operating on the first complex factor gives a different result. We see that

$$\psi_n^* = \psi_n \Rightarrow \psi = \psi^*$$

Thus we can write that

$$\langle E \rangle = \int_0^a \hat{H} \psi \psi dx$$

We now basically have the answer where

$$\begin{aligned} \langle \hat{H} \rangle &= \int_0^a (\hat{H} \psi) \psi dx \\ &= \frac{1}{2} \int_0^a [\hat{H}(\psi_1 + \psi_2)] (\psi_1 + \psi_2) dx \\ &= \frac{1}{2} \int_0^a [E_1 \psi_1 + E_2 \psi_2] (\psi_1 + \psi_2) dx \\ &= \frac{1}{2} \int_0^a [E_1 \psi_1^2 + E_1 \psi_1 \psi_2 + E_2 \psi_1 \psi_2 + E_2 \psi_2^2] dx \\ &= \frac{1}{2} (E_1 + E_2) \int_0^a \psi_1 \psi_2 (E_1 + E_2) dx \end{aligned}$$

Now we wish to consider the generic case

$$\begin{aligned} &\int_0^a \psi_m \psi_n dx \\ &= \int_0^a \frac{2}{a} \sin\left(m\pi \frac{x}{a}\right) \sin\left(n\pi \frac{x}{a}\right) dx \\ &= \frac{2}{a} \int_0^a \left[\cos\left(\frac{m-n}{a}\pi x\right) - \cos\left(\frac{m+n}{a}\pi x\right) \right] dx \\ &= \frac{2}{a} \left[\frac{a}{(m-n)\pi} \sin\left(\frac{m-n}{a}\pi x\right) - \frac{a}{(m+n)\pi} \sin\left(\frac{m+n}{a}\pi x\right) \right] \Big|_0^a \\ &= 0, m \neq n \end{aligned}$$

To summarize, we have shown that

$$\int_0^a \psi_n \psi_m dx = \begin{cases} 1, & \text{if } m = n; \\ 0, & \text{if } m \neq n; \end{cases} = \delta_{mn}$$

therefore the cross terms must go away and will be just the average value.

Remark. Time Dependence:

$$\psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right) e^{-i \frac{E_n}{\hbar} t}$$

What does our superposition look like:

$$\psi = \frac{1}{\sqrt{2}} [\psi_1 + \psi_2] = \frac{1}{\sqrt{2}} \left[\psi_1 e^{-i \frac{E_1}{\hbar} t} + \psi_2 e^{-i \frac{E_2}{\hbar} t} \right]$$

$$\langle \hat{H} \rangle = \frac{1}{2} \int_0^a (\hat{H} \psi^*) \psi dx$$

Note that we can no longer ignore the complex conjugate term and compute the quantity

$$\begin{aligned} \langle \hat{H} \rangle &= \frac{1}{2} \int_0^a \left[\hat{H} \left(\psi_1 e^{\frac{iE_1}{\hbar} t} + \psi_2 e^{\frac{iE_2}{\hbar} t} \right) \right] \left(\psi_1 e^{-\frac{iE_1}{\hbar} t} + \psi_2 e^{-\frac{iE_2}{\hbar} t} \right) dx \\ &= \frac{1}{2} \int_0^a \left[E_1 \psi_1 e^{i\omega_1 t} + E_2 \psi_2 e^{i\omega_2 t} \right] \left[\psi_1 e^{-i\omega_1 t} + \psi_2 e^{-i\omega_2 t} \right] dx \\ &= \frac{1}{2} \int_0^a \left[E_1 \psi_1^2 + E_2 \psi_2^2 + E_1 \psi_1 \psi_2 e^{i(\omega_1 - \omega_2)t} + E_2 \psi_1 \psi_2 e^{-i(\omega_1 - \omega_2)t} \right] dx \\ &= \frac{1}{2} (E_1 + E_2) + \frac{1}{2} \int_0^a E_1 \psi_1 \psi_2 e^{i(\omega_1 - \omega_2)t} + E_2 \psi_1 \psi_2 e^{-i(\omega_1 - \omega_2)t} dx \\ &= \frac{1}{2} (E_1 + E_2) \end{aligned}$$

Remark. When we think about vector spaces we usually think of our own Euclidean space with three basis vectors that can create any vector in the subspace where

$$\vec{r} = a\hat{x} + b\hat{y} + c\hat{z}$$

or written as

$$\sum_{i=1}^3 \alpha_i \hat{x}_i$$

In principle, the superposition state is the sum of two basis functions. Thus, we can write down the wave function in two different ways:

$$\psi = \sum_{n=1}^{\infty} c_n \psi_n$$

now this is our vector spanned by our basis vectors ψ_n

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{a}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Now we realize that quantum mechanics is just linear algebra where r

$$\hat{H}_{\infty \times \infty} \vec{\psi} = \vec{E}$$

Note that all must happen is that we satisfy the boundary conditions and end up with a DFT.

Lecture 5

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Chapter 3

Linear Algebra

Lecture 6

3.1 Algebra Notation

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Recall that the definition of a bra and ket vector where the ket vector is a column vector that is N dimensional and the bra vector is the dual space of the ket vector. We define the inner product as

$$\begin{aligned}\langle a|a\rangle &= \sum_i a_i^* a_i \langle u_j|u_j\rangle \\ &= \sum_i |a_i|^2\end{aligned}$$

We can the normalized vector of $|a\rangle$ to be

$$\frac{|a\rangle}{\sqrt{\langle a|a\rangle}} \Rightarrow \langle a|a\rangle_{norm} = \frac{\langle a|a\rangle}{\langle a|a\rangle}$$

In the context of quantum mechanics we can have two states that we define as

$$|\psi\rangle = A \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

which represent the two states

$$\Rightarrow \langle \psi|\psi\rangle = 1 \Rightarrow A = \frac{1}{\sqrt{\alpha_1^2 + \alpha_2^2}}$$

Recall the square well and how the normalization is present here. Thus our state is

$$|\psi\rangle = \frac{1}{\sqrt{\alpha_1^2 + \alpha_2^2}} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

We write the probability of measuring such as $\sqrt{P_1}$ and $\sqrt{P_2}$. With this notation we can determine what the expectation values are.

3.2 Expectation Values

Suppose we have some operator $\hat{O}$, then the expectation value corresponds to

$$\frac{\langle \psi|\hat{O}|\psi\rangle}{\langle \psi|\psi\rangle}$$

Assuming that the wave function is already normalized, we can compute the numerator

$$\langle \psi|\left[\hat{O}|\psi\rangle\right]$$

$$\begin{aligned}
\hat{O}|\psi\rangle &= \sum_i \alpha'_i |u_i\rangle \\
\langle\psi|\psi'\rangle &= \sum_{i,j} \alpha_j^* \alpha'_i \langle u_j|u_i\rangle \\
&= \sum_j \alpha_j^* \alpha'_j = \langle\hat{O}\rangle
\end{aligned}$$

This property that the operator on the wavefunction spits out a new wavefunction can be represented as a matrix of dim $N \times N$

$$\begin{pmatrix} & \end{pmatrix}_{N \times N} \begin{pmatrix} \end{pmatrix} = \begin{pmatrix} \end{pmatrix}$$

One question to ask is how can we find the variance of our expectation value (probability)?

$$\begin{aligned}
\sigma_{\hat{O}}^2 &= \langle (\hat{O} - \langle\hat{O}\rangle)^2 \rangle \\
&= \langle\psi| (\hat{O} - \langle\hat{O}\rangle)^2 |\psi\rangle
\end{aligned}$$

Expanding out this quantity we can write this as

$$\langle\psi| (\hat{O} - \langle\hat{O}\rangle) (\hat{O} - \langle\hat{O}\rangle) |\psi\rangle = 0$$

Expanding the LHS2 we have

$$\hat{O}|\psi\rangle - \langle\hat{O}\rangle|\psi\rangle$$

the LHS1 is going to be the dual of LHS2. Therefore we can write this as

$$\langle (\hat{O} - \langle\hat{O}\rangle) \psi |$$

This implies that LHS2 must go to 0 where we need to find a vector such that it satisfied

$$\hat{O}|\psi\rangle = \langle\hat{O}\rangle|\psi\rangle$$

Therefore the eigenvalue of the eigenstate is going to be the expectation value of the operator. This is also the case for

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

Example.

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, | -1 \rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Another example is

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\det(\hat{\sigma}_y - \lambda I) = 0$$

$$|1\rangle = \begin{pmatrix} 1 \\ i \end{pmatrix}, | -1 \rangle = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Definition 3.2.1. We have some vector and their properties

$$\begin{aligned}
|\psi\rangle &= \begin{pmatrix} \end{pmatrix} \\
\langle\psi| &= (|\psi\rangle^T)^* = |\psi\rangle^\dagger
\end{aligned}$$

$$\begin{aligned}\left[\hat{O}|\psi\rangle\right]^\dagger &= [|\psi'\rangle]^\dagger \\ \Rightarrow \langle\psi|\hat{O}^\dagger &= \langle\psi'|\end{aligned}$$

Let us try to compute the following

$$\begin{aligned}\langle\psi|\hat{O}\psi\rangle &= \alpha \\ \langle\hat{O}\psi|\psi\rangle &= \langle\psi|\hat{O}^\dagger|\psi\rangle = \alpha^*\end{aligned}$$

For Hermitian operators we have the property that

$$\hat{O} = \hat{O}^\dagger$$

3.3 Commutators

Suppose we had an operator

$$\hat{O}|\psi_N\rangle = \alpha_N|\psi_N\rangle$$

On what condition can we find a second operator that satisfies the following:

$$\hat{Q}|\psi_N\rangle = \beta_N|\psi_N\rangle$$

where different operators have the same eigenvector. We can compute the quantity

$$\hat{Q}\hat{O}|\psi_N\rangle = \beta_N\alpha_N|\psi_N\rangle$$

$$\hat{O}\hat{Q}|\psi_N\rangle = \alpha_N\beta_N|\psi_N\rangle$$

Therefore we must have that

$$\hat{Q}\hat{O} - \hat{O}\hat{Q} = 0$$

$$[\hat{Q}, \hat{O}] = 0$$

where these two operators are simultaneously diagonalizable. Hello